

From pipeline alert to evidence-backed action.

Business proposal, product requirements, and reference architecture for a governed AI agent that investigates data-platform incidents and recommends the next best action.

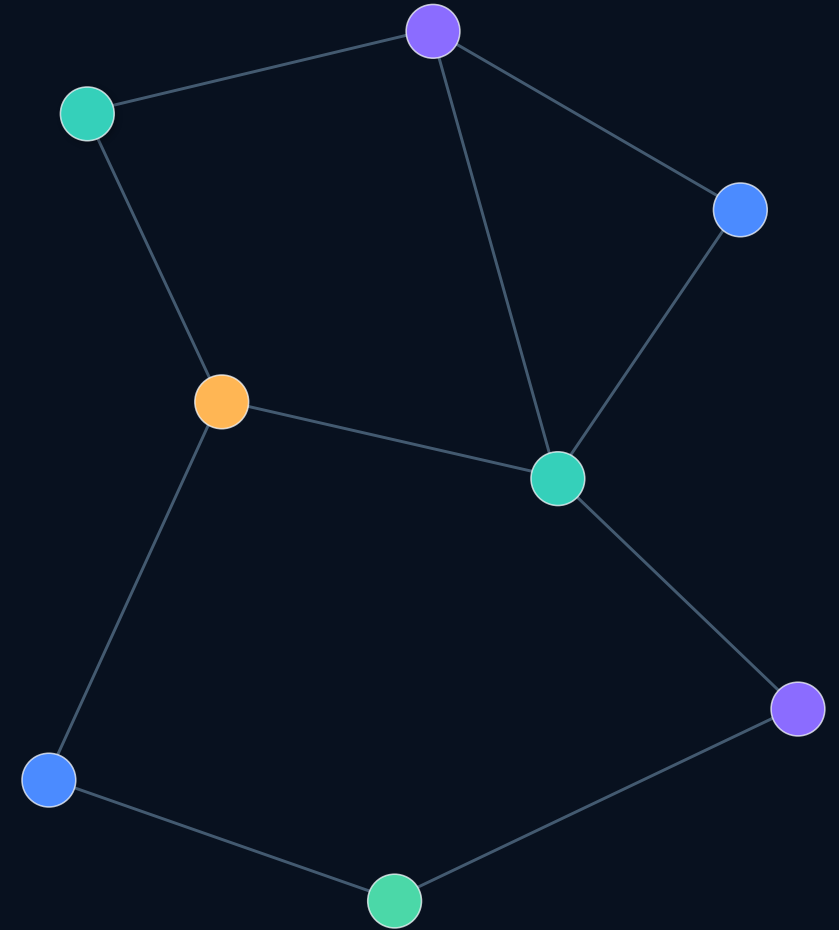
READ-ONLY MVP

DATABRICKS FIRST

PLATFORM-AGNOSTIC

Draft v0.1 | September 2026

Prepared for business, product, engineering, security, and operations stakeholders



The proposal in one slide

Authorize a governed, read-only pilot that proves business value before any autonomous production action.

1

Business problem

Incident diagnosis is fragmented across logs, jobs, lineage, tickets, code, and tribal knowledge.

- Slow time to first diagnosis
- High expert dependency
- Repeated investigations
- Weak evidence and audit trail

2

Proposed product

NEXUS correlates platform evidence, explains likely root cause, and recommends a governed next action.

- Databricks-first investigation
- Evidence-linked conclusions
- Human approval for writes
- Reusable agent service

3

Decision requested

Approve a 12-week MVP pilot with a named sponsor, one pilot domain, and read-only access to required systems.

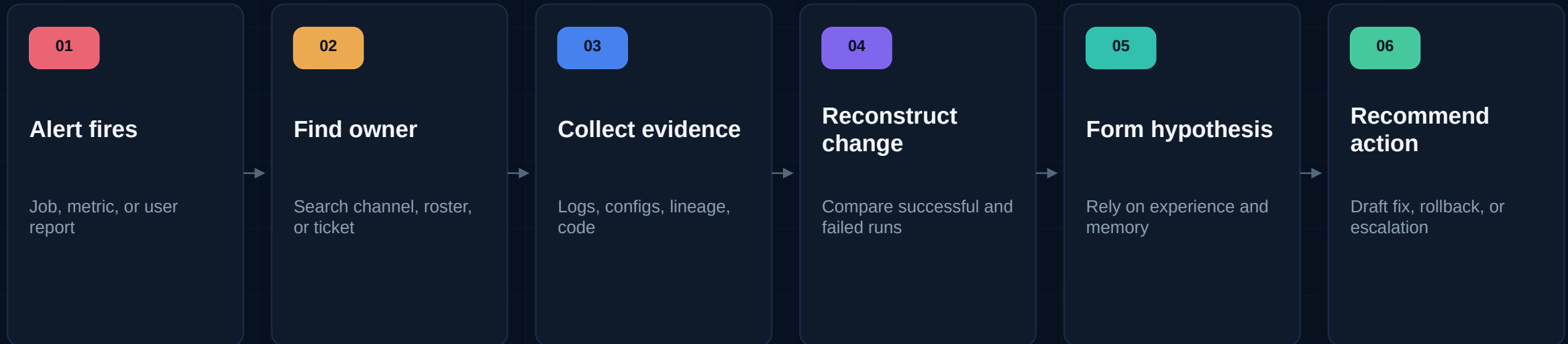
- Target: first useful diagnosis in under 5 minutes
- Target: 50% reduction in median triage time
- No autonomous production change in MVP

All targets are proposed pilot goals and should be baselined against the selected domain before approval.

03 THE OPERATING PROBLEM

Today: six handoffs before a credible answer

Every handoff adds delay, context loss, and dependence on the engineer who already knows the system.



The product opportunity is not to generate a prettier error summary. It is to compress this entire evidence-gathering and reasoning path into a governed, repeatable workflow.

The cost is larger than engineering time

Long investigations affect delivery velocity, service reliability, customer confidence, and control effectiveness.

MTTR

RELIABILITY

Longer incident duration

More time between alert, diagnosis, and safe recovery.

FOCUS

PRODUCTIVITY

Senior engineer interruption

High-value specialists become the default search engine.

REUSE

KNOWLEDGE

Tribal resolution history

Past incidents are hard to reuse consistently across teams.

TRUST

GOVERNANCE

Weak evidence chain

Decisions and access are difficult to reconstruct later.

The pilot should baseline both direct engineering effort and reliability outcomes; avoided business impact is tracked separately.

One trusted path from signal to decision

NEXUS becomes the AI reliability layer between engineers and the systems they operate.



Four users, one shared operating picture

The same evidence graph supports different decisions without giving every persona the same permissions.

DE

Data engineer

JOB TO BE DONE

Diagnose why a job failed and identify the safest next action.

SUCCESS

Less log hunting; faster root cause.

PO

Platform owner

JOB TO BE DONE

Spot recurring failure patterns and systemic reliability gaps.

SUCCESS

Better prioritization and prevention.

OL

Operations lead

JOB TO BE DONE

Understand impact, confidence, owner, and recovery status.

SUCCESS

Clearer incident command decisions.

SA

Security / audit

JOB TO BE DONE

Verify what data and tools were accessed, by whom, and why.

SUCCESS

Policy enforcement and traceability.

A decision workspace, not another chat window

The answer is useful only when the user can inspect evidence, confidence, impact, and the proposed action.

nexus.internal / incidents / INC-1842

INCIDENTS

INC-1842

customer_daily_ingestion

INC-1839

payment_stream_lag

INC-1835

recon_schema_drift

Why did customer_daily_ingestion fail?

DIAGNOSIS READY

N

Likely root cause

Upstream column customer_type changed from STRING to STRUCT without a compatible ingestion rule.

Confidence 0.92 | First failure 03:05 UTC | Last successful run 02:05 UTC

VIEW EVIDENCE

DRAFT REMEDIATION

ESCALATE

EVIDENCE

JOB RUN

Task 2 failed: schema mismatch

SCHEMA

customer_type STRING -> STRUCT

LINEAGE

3 downstream tables affected

CODE

No related change in current release

Every conclusion is linked to inspectable evidence.

Five product principles that shape every design choice

The MVP is successful only when users trust both the conclusion and the boundaries of the system.

01 Evidence-linked

No unsupported root-cause claim. Show the artifacts used and the gaps that remain.

02 Identity-aware

The agent never gains more access than the invoking user or an explicitly approved service role.

03 Read-only first

Investigation is autonomous; changes, deployments, and data writes require separate policy and approval.

04 Uncertainty visible

Confidence, alternative hypotheses, missing context, and stale evidence are surfaced explicitly.

05 Auditable by design

Prompts, plans, tool calls, data access, decisions, approvals, and outcomes form one trace.

Prove investigation value before introducing execution risk

The first release is intentionally narrow: one domain, one primary platform, evidence-driven analysis, and no production writes.

IN MVP

Read-only incident investigator

- Manual prompt and Databricks job-failure trigger
- Job, task, cluster, log, and run-history inspection
- Schema and lineage context where authorized
- Runbook and past-incident retrieval
- Root-cause hypotheses with confidence and evidence
- Recommended next action and escalation path
- Full trace, access log, and outcome feedback

OUT OF MVP

Deferred until value and controls are proven

- Autonomous code changes or pull requests
- Production deployment, rollback, or job restart
- Cross-domain unrestricted access
- Automated incident closure
- Multi-agent debate or self-directed long-running tasks
- Predictive anomaly detection as a core feature
- Replacement of existing monitoring or ITSM platforms

What the MVP must do - and how well it must do it

This is the contract between the business outcome, the product behavior, and the engineering implementation.

Core capabilities

- 1 Accept incident question or event
- 2 Resolve job and run context
- 3 Select and call approved tools
- 4 Build evidence graph
- 5 Generate ranked hypotheses
- 6 Recommend next action

Trust and control

- 1 User and tenant identity
- 2 Per-tool authorization
- 3 PII-aware retrieval and redaction
- 4 Read-only enforcement
- 5 Confidence and citations
- 6 Immutable audit trace

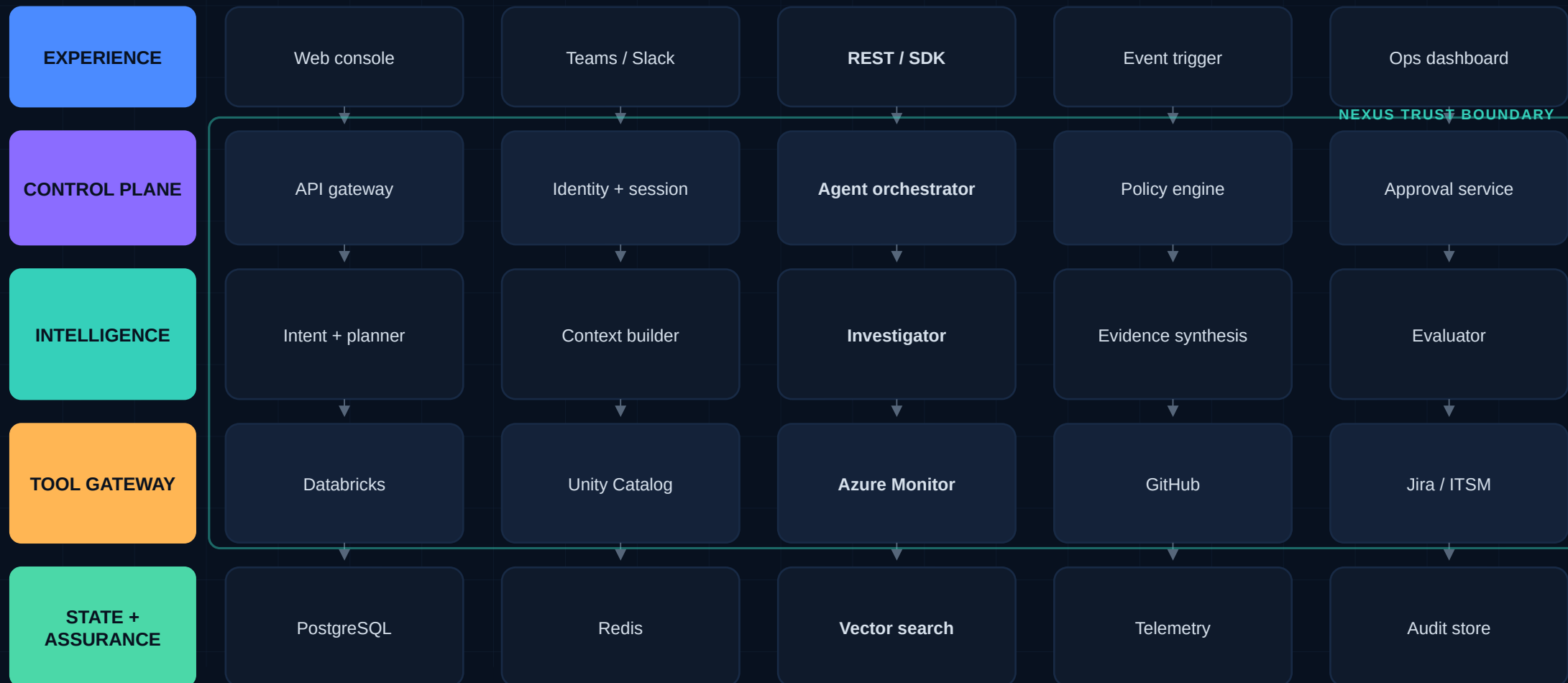
Operational quality

- 1 Useful diagnosis ≤ 5 minutes
- 2 Graceful partial-tool failure
- 3 Retry and timeout controls
- 4 Observable token and tool cost
- 5 Versioned prompts and policies
- 6 Feedback and evaluation loop

Detailed user stories and acceptance criteria are included in the appendix.

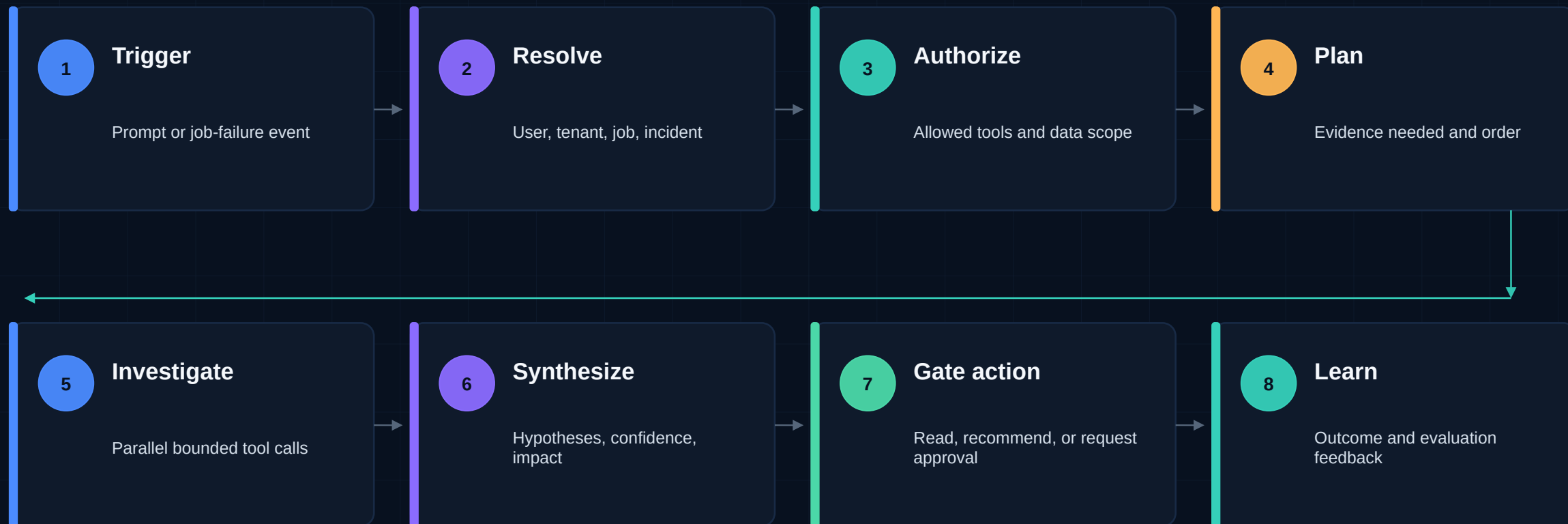
NEXUS is an independent agent service with governed connectors

Databricks is Tool #1 - not the hosting boundary. The same agent can serve web, Teams, APIs, and event-driven workflows.



How a request becomes an evidence-backed decision

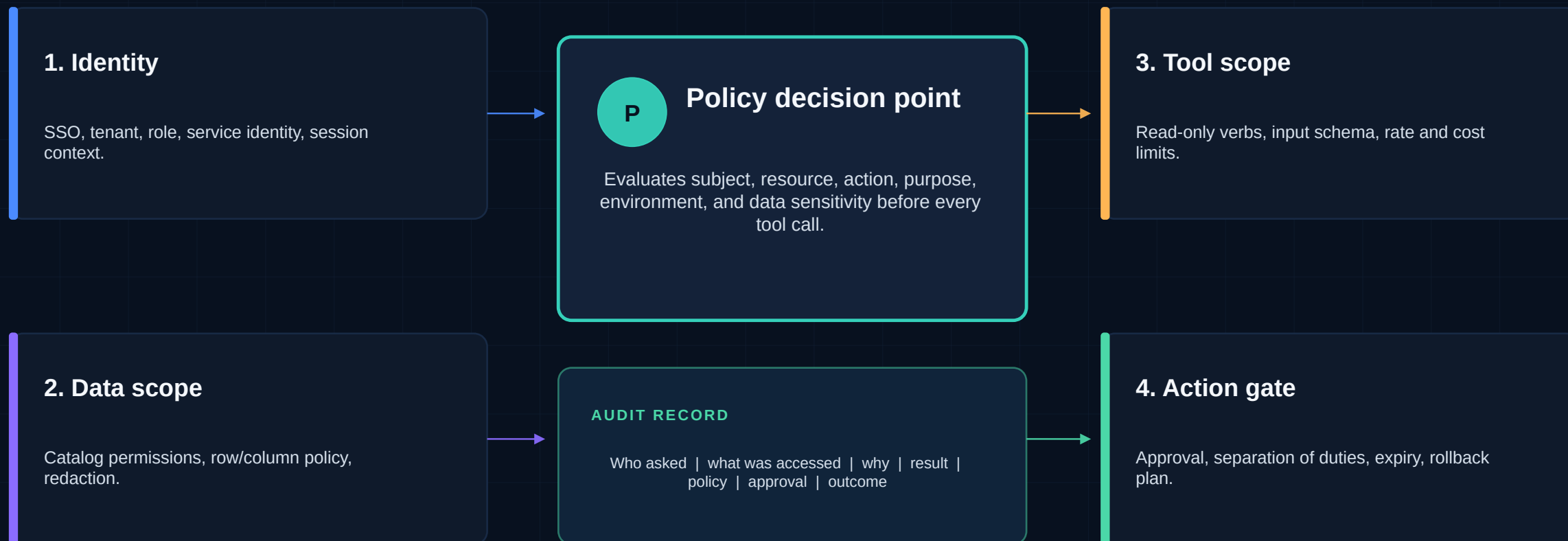
Planning and tool execution are bounded by identity and policy at every step - not checked only at the end.



Each tool call is time-bound, cost-bound, input-validated, and recorded in the trace.

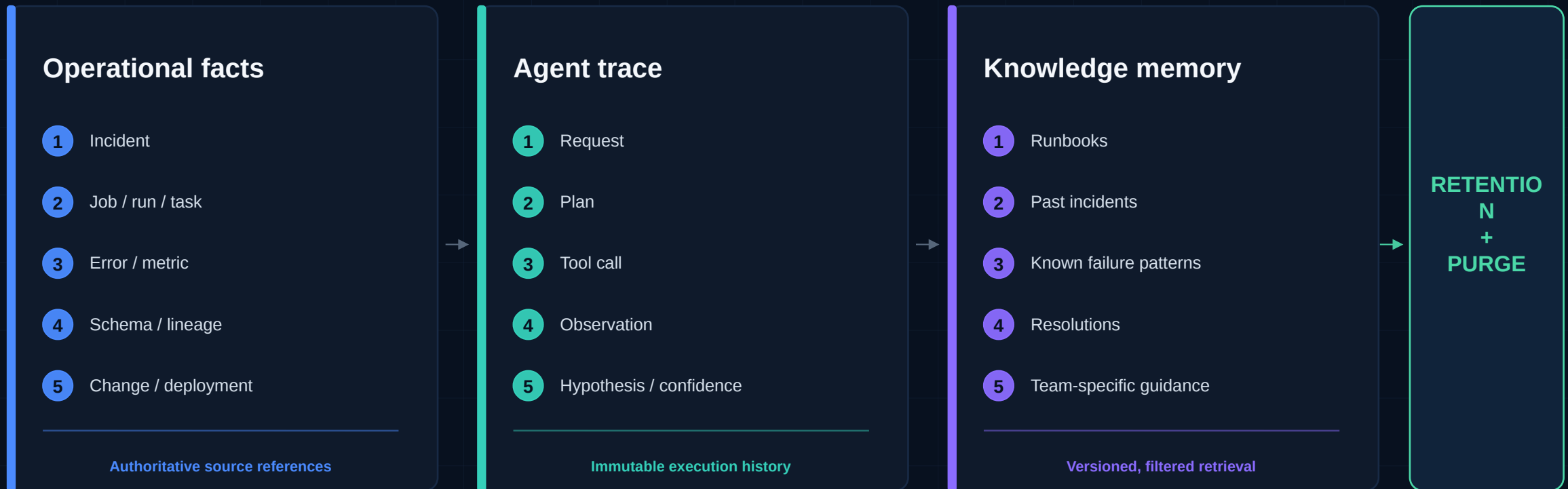
The agent is powerful only inside explicit policy boundaries

Identity, data authorization, tool permissions, and approval policy are independent controls.



Separate system-of-record facts from probabilistic agent memory

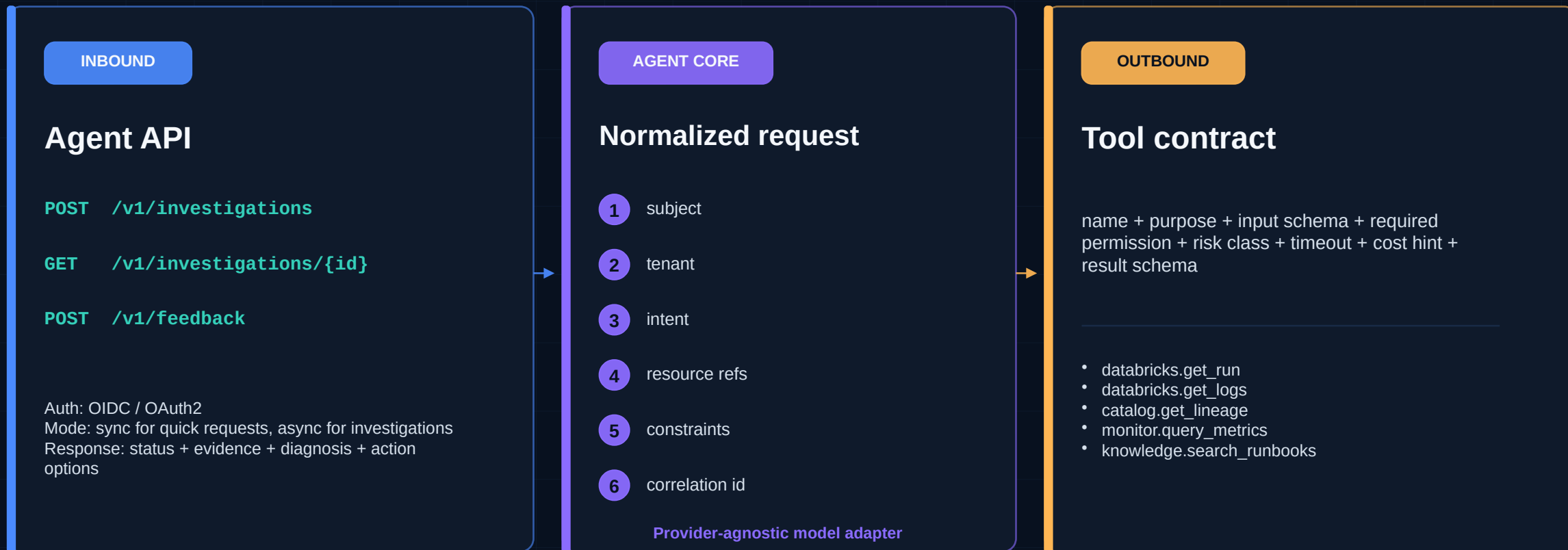
Reliable agents need a structured incident model, an evidence graph, and tightly governed retrieval - not one unbounded vector store.



Recommended default: retain raw tool payloads for the minimum operational and regulatory period; retain derived summaries only while still useful and authorized.

A portable agent requires stable contracts at every boundary

Channels call the agent service; the agent calls tools through a normalized, policy-aware gateway.



A 12-week path to a controlled pilot

The plan front-loads access, evaluation, and security decisions so the team does not discover them at the end.



Core team

Product owner | AI/platform architect | 2 agent/backend engineers | data-platform engineer | fractional security, UX, SRE/QA

Pilot timing assumes timely access to Databricks APIs, logs, selected catalog metadata, runbooks, and representative incident history.

17 SUCCESS MEASURES

A pilot should prove speed, quality, trust, and economics

Usage alone is not success. The product must improve a real reliability outcome without weakening control.

< 5 min

Time to first useful diagnosis

Proposed MVP target

-50%

Median human triage time

Versus pilot baseline

>= 90%

Conclusions linked to evidence

Coverage target

100%

Tool calls policy-checked and audited

Control target

Value model

Direct monthly productivity value

=

Incidents / month

X

Minutes saved /
incident

X

Loaded hourly cost

/

60

Risk avoidance, faster recovery, fewer repeat incidents, and better auditability are tracked separately and may dominate the direct labor value.

Pilot decision rule

- Scale when speed and quality targets are met
- Pause if access or evidence controls are unreliable
- Redesign before enabling any write action

The largest risks are product and control risks - not model selection

The MVP should make these decisions explicit before implementation hardens around hidden assumptions.

False confidence

Evidence coverage threshold, alternatives, abstention, evaluation set.

Over-broad access

Identity passthrough, catalog policy, per-tool authorization, redaction.

Tool or log gaps

Partial-result behavior, stale-data warnings, explicit unknowns.

Cost / latency drift

Budgets, timeouts, parallelism limits, caching, smaller models.

Low engineer trust

Inspectable evidence, feedback loop, shadow mode, named owners.

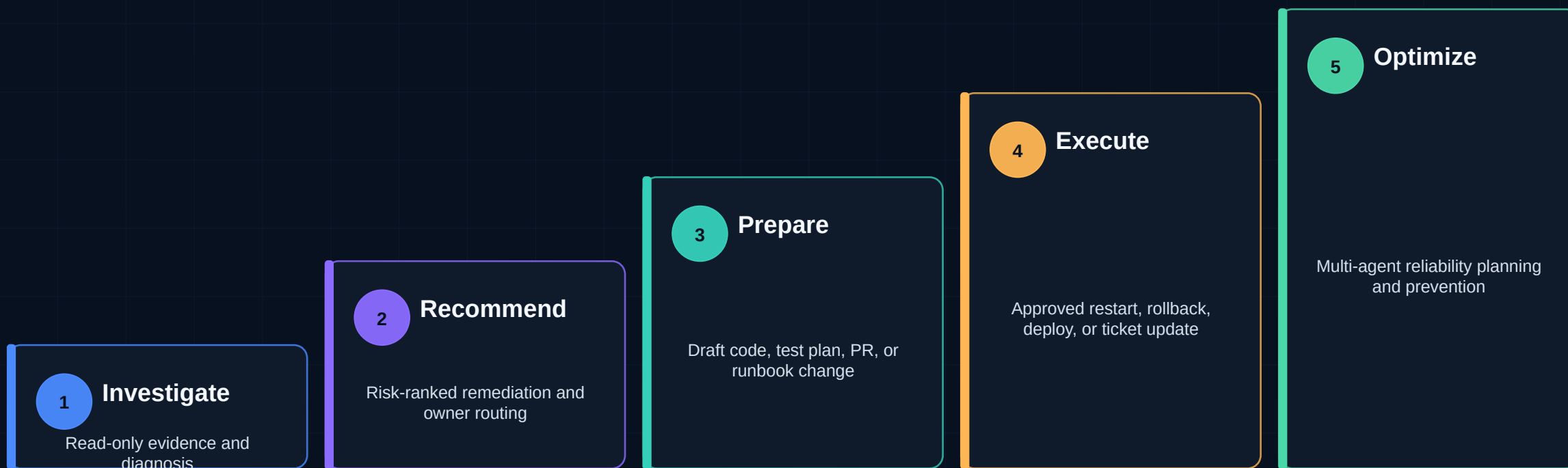
Scope expansion

Read-only MVP boundary and a formal gate for remediation features.

Decisions required for kickoff: pilot domain | data residency | model hosting posture | retention | access model | success baseline | business sponsor

From investigator to governed engineering operating system

The architecture is designed to add capabilities gradually without turning the first pilot into an autonomous production risk.



Autonomy increases only after evidence quality, access control, and approval workflows meet explicit gates.

DECISION

Approve the NEXUS MVP pilot

A read-only, Databricks-first AI investigator that proves measurable reliability value while establishing the governance foundation for future agentic engineering.

- 1 Name a business sponsor and pilot domain
- 2 Approve read-only access and security discovery
- 3 Baseline incidents, triage time, and quality
- 4 Fund a 12-week cross-functional MVP team

PILOT EXIT

One evidence-based scale decision

- Did diagnosis become materially faster?
- Were conclusions accurate and inspectable?
- Did policy controls work under real use?
- Is the economics case strong enough to scale?
- Which capability should be added next?

NEXUS

Priority user stories for MVP

These stories translate the proposal into buildable, testable product behavior.

ID	USER STORY	PRIORITY
US-01	As a data engineer, I can ask about a failed job using a natural-language question or run ID.	P0
US-02	I can see the exact evidence used for each diagnosis and open the source artifact.	P0
US-03	I can see ranked root-cause hypotheses, confidence, missing context, and alternative explanations.	P0
US-04	The agent uses only tools and data that I am authorized to access.	P0
US-05	I can view a recommended next action, risk level, owner, and escalation path.	P0
US-06	I can provide feedback on correctness and record the actual resolution.	P1
US-07	A job failure event can create an investigation automatically in shadow mode.	P1
US-08	An auditor can reconstruct the complete request, access, tool, policy, and outcome trace.	P0

Minimum acceptance criteria and non-functional gates

A feature is not done until the user outcome, control behavior, and operational telemetry are all verifiable.

Diagnosis quality

- At least one ranked hypothesis or explicit abstention
- Evidence references open successfully
- Missing context and confidence are visible

Authorization

- Denied resources are never retrieved or summarized
- Every tool call includes subject and purpose
- Read-only methods enforced at gateway

Resilience

- One failed tool does not fail the entire investigation
- Timeouts and retries are bounded
- Partial and stale evidence is labeled

Observability

- Trace spans request, plan, tools, model, policy, response
- Token, latency, and tool-cost metrics captured
- Audit record exportable by incident ID

Performance

- Initial acknowledgement ≤ 2 seconds
- Useful diagnosis target ≤ 5 minutes
- Concurrency target defined from pilot load

Product safety

- No production write path in MVP
- Prompt injection tests included
- Sensitive output redaction validated

Open decisions to close in discovery

These choices should be documented as architecture decision records before build commitments.

D-01

Pilot domain

Which team and incident class provide enough volume, value, and willing users?

D-02

Hosting and residency

Where do the agent runtime, model endpoint, trace, and memory reside?

D-03

Identity model

User delegation, service role, or hybrid - and how is purpose propagated?

D-04

Knowledge scope

Which runbooks, incidents, and code metadata can be indexed and retained?

D-05

Evaluation truth set

Who labels root cause, evidence quality, and acceptable abstention?

D-06

Model posture

Provider abstraction, approved model classes, fallback, and data-use terms?

D-07

Retention

How long are raw tool payloads, traces, summaries, and feedback retained?

D-08

Scale gate

What exact thresholds authorize expansion to PR generation or approved actions?

Recommended governance: one accountable product owner, one architecture decision log, one risk register, and one measurable pilot scorecard.